

Public Companies: Semiconductor Packaging

Advanced packaging is now the primary bottleneck in AI chip supply, ahead of transistor fabrication itself

Context: CoWoS (TSMC's leading advanced packaging tech) is sold out through 2026. TSMC is scaling CoWoS capacity from ~35,000 to ~130,000 wafers/month, still not enough to meet demand. Global CoWoS demand is projected to reach ~1 million wafers in 2026, up from 370,000 in 2024. Memory and packaging together now represent 60-70% of AI accelerator cost -- logic silicon is no longer the dominant cost driver.

1. Foundries & IDMs with Captive Advanced Packaging

The largest players run advanced packaging in-house rather than outsourcing -- this is where the actual CoWoS/EMIB/Foveros bottleneck lives.

Company	Ticker	Role	Recent Signal (2026)
TSMC	TSM	Owns CoWoS, InFO, and SoIC -- the dominant advanced packaging platforms; every leading AI GPU depends on its capacity	Scaling CoWoS from ~35K to ~130K wafers/month by end of 2026; still sold out; new Chiayi AP7 and Arizona packaging plants
Samsung Electronics	KRX: 005930	I-Cube (2.5D) and X-Cube (3D) advanced packaging; integrated with its own HBM production	Developing 3.3D advanced packaging targeting Q2 2026 mass production for AI chips
Intel	INTC	EMIB and Foveros advanced packaging, positioned as an alternative to TSMC for US-based capacity	Cited as a potential US CoWoS alternative given domestic packaging capacity in New Mexico, Oregon, Arizona

2. OSAT -- Outsourced Semiconductor Assembly & Test

Handles packaging/test for fabless customers and overflow capacity from foundries. Top 3 (ASE, Amkor, JCET) hold 60%+ of global OSAT revenue.

Company	Ticker	Role	Recent Signal (2026)
ASE Technology Holding	ASX / TWSE: 3711	World's largest OSAT; VIPack platform; critical CoWoS backend processing partner to TSMC	Investing \$200M in large-format panel-level packaging for AI chips; advanced packaging sales projected to double in 2026
Amkor Technology	AMKR	Largest US-headquartered OSAT; flip chip, wafer-level, and TSV packaging	\$2B Arizona advanced-packaging campus (co-located with TSMC Phoenix, \$400M CHIPS Act support); 2026 capex guidance \$2.5-3B, 30-35% toward advanced packaging/test
JCET Group	SSE: 600584	Largest China-based OSAT; XDFOI heterogeneous integration chiplet technology	Revenue grew 19.3% in the most recent period on advanced packaging/automotive demand
Powertech Technology	TWSE: 6239	Leading memory packaging OSAT, key for HBM stacking	Direct beneficiary of HBM stacking demand growth alongside SK Hynix/Samsung/Micron ramp
Tongfu Microelectronics	SZSE: 002156	China OSAT with deep strategic partnership with AMD for CPU/GPU packaging	Rapid growth benefiting from AMD data center chip packaging relationship

ChipMOS Technologies	IMOS	Display driver IC and memory test specialist	Niche exposure to memory test volume growth
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3. Advanced Packaging & Bonding Equipment

Hybrid bonding and thermal compression bonding tools are the key equipment bottleneck behind the OSATs/foundries -- concentrated among a handful of suppliers.

Company	Ticker	Role	Recent Signal (2026)
BE Semiconductor Industries (Besi)	AMS: BESI	Market leader in hybrid bonding and thermo-compression bonding equipment	Rising bookings for HBM4/leading-edge logic hybrid bonding tools; Applied Materials took a 9% strategic stake (2025) to deepen collaboration
ASMPT	HKEX: 0522	Die bonders, wire bonders, and advanced packaging assembly equipment	Core equipment supplier alongside Besi and K&S for AI packaging capacity expansion
Kulicke & Soffa	KLIC	Die/wire bonders and advanced packaging equipment	Benefiting from OSAT advanced packaging capacity expansion cycle through 2026
Applied Materials	AMAT	Broad semiconductor equipment maker with growing advanced packaging/hybrid bonding tools	Took equity stake in Besi to co-develop hybrid bonding tech for advanced packaging
DISCO Corporation	TYO: 6146	Dicing and grinding equipment essential for die singulation in advanced packaging	Named among core advanced packaging equipment suppliers benefiting from AI packaging ramp
TOWA Corporation	TYO: 6315	Molding and packaging equipment	Positioned alongside DISCO as a core equipment beneficiary of packaging capacity expansion

4. Semiconductor Test Equipment

Advanced packages require substantially more test complexity (Known Good Die testing, interposer yield checks) than monolithic chips, adding 15-30% to test cost.

Company	Ticker	Role	Recent Signal (2026)
Teradyne	TER	Leading automated test equipment (ATE) maker for advanced packaging/AI chips	Direct beneficiary of rising test complexity from 2.5D/3D packaging and chiplet architectures
Advantest	TYO: 6857	Major ATE supplier, especially strong in memory and SoC test	Called out for ~40% YoY equipment increase tied to HDFO and test capacity expansion (2026)
Cohu	COHU	Test handling and interface equipment	Smaller-cap beneficiary of rising test intensity for advanced-packaged AI chips

5. Advanced Substrates & Interposer Materials

Ajinomoto Build-up Film (ABF) substrates are near-sole-sourced and were in short supply through 2025; 12-layer substrate lead times stretched to 38 weeks.

Company	Ticker	Role	Recent Signal (2026)
Ibiden	TYO: 4062	Leading high-layer-count ABF substrate manufacturer	Capacity expansion targeted to reach volume by mid-2026 to relieve substrate shortage
Shinko Electric Industries	TYO: 6967	Major advanced substrate manufacturer alongside Ibiden	Co-leads Japan's substrate capacity buildout amid continued shortage into 2026
Unimicron	TWSE: 3037	Taiwan-based substrate and PCB manufacturer	Key Taiwan-based substrate supplier to the AI packaging supply chain
Samsung Electro-Mechanics	KRX: 009150	Substrate manufacturer, part of the Samsung ecosystem	Supplies substrates supporting Samsung's advanced packaging and HBM integration

6. HBM Memory (Packaging-Adjacent)

Not packaging companies themselves, but HBM stacking is co-integrated with advanced packaging and is equally supply-constrained -- excluding them would misrepresent the bottleneck.

Company	Ticker	Role	Recent Signal (2026)
SK Hynix	KRX: 000660	Supplies majority of HBM for Nvidia's Blackwell-class GPUs; packages its own HBM stacks captively	Capacity-constrained through 2026 despite being the leading HBM supplier
Micron Technology	MU	Ramping HBM4 for volume production	Cited \$25B investment framed as a bet on AI memory as the next growth driver
Samsung Electronics	KRX: 005930	Also a major HBM supplier alongside its packaging business (see Category 1)	Repricing HBM contracts upward in the high-teens percentage range on tight supply (2026)

Categories/Notes Worth Adding

- **China-domestic OSAT/localization plays** — Tianshui Huatian (HT-Tech) and other Chinese OSATs are growing 2-3x faster than the global market on China's semiconductor self-sufficiency policy; a distinct, higher-geopolitical-risk sub-theme from the Western/Taiwan names above.
- **Glass substrate/glass-core packaging** — an emerging next-generation substrate technology (beyond silicon interposers) that ASE and Amkor have begun pilot lines for; still pre-commercial but flagged by multiple 2026 sources as the next technical inflection point.
- **Panel-level packaging (PLP)** — an alternative to wafer-level packaging for large-format AI chips; Silicon Box (private, EU-backed) and ASE's large-format panel investment are early movers here.
- **Materials suppliers** (mold compounds, underfill, thermal interface materials) — e.g. Sumitomo Bakelite, Resonac (Showa Denko) -- a thinner, more specialized layer not broken out separately here but adjacent to the substrate category above.

Generated July 10, 2026. This is a disjoint reference set, not a ranked or filtered list.